Sebastian Kobler

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SUMMARY

PhD researcher at the University of Sydney working at the intersection of Natural Language Processing and education. My research focuses on how people interact with LLM-based systems and how to measure whether those interactions achieve their intended goals. I develop computational frameworks for evaluating human-AI dialogue, with published work at ACL 2026. More broadly, I am interested in building NLP systems that support effective human-AI collaboration, particularly in high-stakes domains. I build generalisable NLP methods in applied settings where the gap between system design and real-world usage has measurable consequences.

PUBLICATIONS

- **S. Kobler**, M. Clemson, A. Sun, J. Kummerfeld. **Your Students Don't Use LLMs Like You Wish They Did.** ACL 2026 (Main Conference). **Score of 10 - Seminal Paper - Top 5% of Accepted Papers.**
- Introduced six computational metrics for evaluating pedagogical alignment in student-AI dialogue. Analysed 12,650 messages across 500 conversations from five AI tutoring systems spanning four university courses. Identified a fundamental usage gap between educator design intent and actual student behaviour.

- **S. Kobler**, J. Kummerfeld. **Off-Topic Detection in Educational LLM Tutors: From Synthetic Stress-Testing to Real-World Validation.** Under review (ACL Rolling Review).
- Evaluated off-topic detection methods across synthetic and real-world settings using 4,983 authentic student conversations. Demonstrated that RAG worsens classification and proposed configurable prompt-based defences.

EDUCATION

University of Sydney - PhD, Computer Science *2024 – Present*
- Focus: NLP for education - evaluating and improving LLM-based educational dialogue systems.
- Supervisor: Dr Jonathan Kummerfeld.

University of Sydney - B.Sc. / B.Adv.Studies (Advanced) *2018 – 2022*
- Majors: Computer Science, Mathematics. First Class Honours. Distinction Average WAM.

RESEARCH PROJECTS

Productive Struggle in AI Tutoring Conversations *2026 – Present*
- Developing a three-way taxonomy (productive struggle, unproductive struggle, answer extraction) for classifying student responses to confusion in AI tutoring.
- Built a two-stage LLM pipeline analysing 3,500 conversations across five platforms, finding that only 37% of confused students engage in productive struggle.
- Validated that productive struggle is situational rather than trait-based, with linguistic markers (self-corrections, hypothesis testing) distinguishing categories.

EduView - Interactive Educational Analytics Dashboard *2025 – Present*
- Designed and built an adaptive AI-powered case study platform (Flask, Socket.IO, OpenAI API) that uses LLMs to generate personalised scenarios and evaluate free-text student responses against learning objectives in real time.
- Implemented a reinforcement learning agent (Q-learning) for dynamic difficulty adjustment based on a 16-dimensional student performance vector during assessment.
- Applied semantic clustering via OpenAI embeddings to aggregate individual knowledge gaps into cohort-level learning analytics for tutors and lecturers. Preparing for classroom pilot deployment.

Differential Privacy and Synthetic Data for Adversarial Robustness (Honours Thesis) *2022*

- Investigated the use of differential privacy and synthetic data generation to defend machine learning models against adversarial attacks. Supervised by Dr Clément Canonne.
- Utilised PyTorch, TensorFlow, and the Synthetic Data Vault to systematically evaluate defence parameters across attack scenarios.

TECHNICAL SKILLS

- **Languages:** Python (primary), C/C++, Java, MATLAB, R
- **ML / NLP:** PyTorch, TensorFlow, Hugging Face Transformers, spaCy, NLTK
- **Tools & Platforms:** OpenAI API, Flask, Socket.IO, SQLAlchemy, AWS (EC2), Docker, Git
- **Research Methods:** Large-scale dialogue analysis, LLM-as-judge evaluation, computational metrics design, RAG systems, prompt engineering

RESEARCH INTERESTS

Human-AI dialogue evaluation, LLM behaviour analysis, computational metrics for interaction quality, adaptive dialogue systems, reinforcement learning for personalisation, NLP for education and learning analytics. Methods developed for educational contexts generalise to evaluating any human-AI dialogue system.

TEACHING EXPERIENCE

Tutor - COMP4446/COMP5046: Natural Language Processing, University of Sydney *2025 – Present*

- Tutor for the undergraduate and postgraduate NLP course covering statistical NLP, deep learning for NLP, and core tasks including parsing, NER, and text classification.

Academic Tutor - Prime Tutoring *2018 – 2021*

- Tutored Mathematics Extension II, Mathematics Extension I, Mathematics, Physics, and Economics at secondary level.